College of Technology Galle

Course: Full Stack Software Development — Object Oriented Programming — Module K72T004M03 — NVQ Level 05

SEMESTER MODEL PAPER 01

Focus: Basic Java, variables, decisions, simple methods & classes

Difficulty: Easy

Duration: 3 Hours **Total Marks:** 100 **Questions:** 7 structured questions (answer any **5**) **Each question:** 20 marks

Instructions to Candidates

1. Answer any **five (5)** questions out of the seven provided.
2. Each question carries **20 marks**. Maximum mark for this paper is **100**.
3. Write answers in the answer booklet. Do **not** write answers on this question paper.
4. Where Java code is required, write clear, syntactically correct Java as taught in this module (VS Code / JDK).
5. Show workings / short explanations where asked. Manage your time (about 35 minutes per question).
6. Mobile phones and programmable devices are not allowed.

Question 1 [20 Marks]

This question covers JDK tools and the structure of a simple Java program.

a. What do the abbreviations **JDK** and **JVM** stand for? State one role of each. **(6 marks)**

b. Name the IDE allowed in this module and the extension pack needed for Java. **(4 marks)**

c. Write the two terminal commands to compile and run a file named `Hello.java` (class `Hello`). **(4 marks)**

d. Why must the `main` method be present in a console application? **(6 marks)**

Question 2 [20 Marks]

Data types and console output.

a. List five primitive Java types used in this module and give one example value for each. **(5 marks)**

b. Write declarations (with initial values) for: student name, age, course fee, paid status, and grade letter. **(5 marks)**

c. Write one `System.out.println` statement that prints: *Hello Amal, age 21* using variables `name` and `age`. **(4 marks)**

d. Explain the difference between `String` quotes and `char` quotes with one example each. **(6 marks)**

Question 3 [20 Marks]

```
int marks = 65;
if (marks >= 50) {
    System.out.println("Pass");
} else {
    System.out.println("Fail");
}
```

Study the decision structure below.

a. What is printed when `marks` is 65? What is printed when `marks` is 40? **(4 marks)**

b. Rewrite the program using an `if / else if / else` chain that prints Distinction (>=75), Credit (65-74), Pass (50-64), or Fail. **(10 marks)**

c. Explain in one sentence when the `else` block runs. **(6 marks)**

Question 4 [20 Marks]

Loops - basics.

a. Write a `for` loop that prints the numbers 1 to 5 (each on a new line). **(6 marks)**

b. Write a `while` loop that prints "Hello" exactly three times. **(6 marks)**

c. State one difference between `while` and `do-while`. **(4 marks)**

d. Why can an infinite loop be a problem in a console menu? **(4 marks)**

Question 5 [20 Marks]

Methods - simple reuse.

a. What does the keyword `void` mean in a method header? **(4 marks)**

b. Write a method `public static void greet(String name)` that prints *Welcome, <name>*. **(6 marks)**

c. Write a method `public static int square(int n)` that returns `n * n`. Show one call from `main`. **(6 marks)**

d. Explain the difference between a parameter and an argument. **(4 marks)**

Question 6 [20 Marks]

Classes and objects - first look.

a. In one sentence each, define **class** and **object**. **(6 marks)**

b. Write a class `Book` with two public fields: `title` (String) and `pages` (int). **(6 marks)**

c. In `main`, create one `Book` object, set both fields, and print them. **(6 marks)**

d. Why can many objects be created from the same class? **(2 marks)**

Question 7 [20 Marks]

Encapsulation - introductory idea.

a. What does encapsulation mean in simple words? Give one real-world analogy (e.g. ATM). **(6 marks)**

b. Why is a public `balance` field dangerous in a bank account class? **(6 marks)**

c. Write a private field `balance` and a public method `getBalance()` that returns it. **(6 marks)**

d. What access modifier means "visible only inside the same class"? **(2 marks)**

******* END OF QUESTION PAPER *******